COURSE INFORMATION

Faculty:	Computing	**Page:**	1 of 6
Program name:	Bachelor in Software Engineering with Honors		
Course code:	SCST1143	**Academic Session/Semester:**	2024/2025 - 1
Course name:	Database Engineering	**Pre/co requisite (course name and code, if applicable):**	-
Credit hours:	3		

Course synopsis	This course introduces students to the concept of database system and how it is used in daily human life and profession. The focus of the course is to equip students with the knowledge and skills on important steps and techniques used in developing a database, especially in the conceptual and logical database design phase. Among topics covered are database environment, database design, entity relationship diagram, normalization, and structured query language (SQL). Students will be taught to use a database management system (DBMS). Students are required to design and develop the database component of an information system using the learned techniques, DBMS and a development tool. At the end of the course, students should be able to apply the knowledge of designing and developing a good database system.			
Course coordinator (if applicable)	Dr Noor Hidayah Zakaria			
Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Rozilawati Binti Dollah			rozilawati

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO*	PLO (Code)	**Taxonomies and ***generic skills	T&L methods	****Assessment methods
CLO1	Design database solutions using the fundamental database principles for the assigned problem according to techniques, DB language, DBMS and development tools.	PLO2 (CG)	C4	Lecture, Active learning	A, T, F
CLO2	Construct databases using a combination of suitable components including database design, back-end framework (Structured Query Language) and use of the security and cloud functions.	PLO3 (PS)	C5	Lecture, Active learning, Project-based learning	A, Q, T, F, PR

Prepared by: Name: Rozilawati Binti Dollah Signature: Date:	Certified by: Name: Signature: Date:

CLO3	Establish good relations among group's leader and members and understand any interchanging role and take responsibilities in executing the group project.	PLO8 (LAR)	TW1, TW2	Project-based learning	PR

**** T – Test; Q – Quiz; A – Assignment; PR – Project; F – Final Exam

Details on Innovative T&L practices:

No.	Type	Implementation
1.	Active learning	Conducted through in-class activities
2.	Project-based learning	Conducted in a group project assignment. Students, in a group of 3 or 4, are required to develop a database application system for a selected company. Students are required to go through the database system development methodology starting from determining the database requirements until developing the database application system. Students shall apply all relevant knowledge acquired throughout the semester on the project. Students are assessed on their ability to deliver a creative database design solution that will be able fulfill the requirements identified.

Weekly Schedule:

Week 1	**Topic 1: Introduction to Database** - File Based Approach - Database Approach - Roles in DB Environment - Database Management System - The 3-Level ANSI-SPARC Architecture
Week 2	**Topic 2: The Relational Model and DB Design** - Terminologies - Integrity Constraints - Views - Database Design Phases - Conceptual Design – Steps & Deliverables
Week 3	**Topic 3: Relational Algebra** - Unary Operations - Set Operations - Join Operations - Division Operation - Aggregation & Grouping Operations
Week 4	**Topic 4: Entity-Relationship (ER) Modeling** - Concepts of ERM (entity, relationships, attributes) - Structural constraint (1:1, 1:*, *:*) - Constraints (Cardinality and Participation)

Week 5	**LAB: SQL - DDL** - Introduction to a DBMS software - SQL: DDL (Data Definition Language) • SQL Data Types • Integrity Enhancement • Data Definition (CREATE, ALTER, DROP) • Views (CREATE, DROP, WITH CHECK OPTION)
Week 6	**LAB: SQL – DDL *(cont.)*** - SQL: DDL (Data Definition Language) • Data Definition (CREATE, ALTER, DROP) • Views (CREATE, DROP, WITH CHECK OPTION) **LAB: SQL- DML (Part 1)** - Database Updates (INSERT, UPDATE, DELETE)
Week 7	**Topic 5: Enhance ER Modeling** - Generalization, Specialization - Superclass, subclass
MID SEMESTER BREAK	
Week 9	**Topic 5: Enhance ER Modeling (cont.)** - Generalization, Specialization - Superclass, subclass **LAB: SQL- DML (Part 2)** - Grouping (GROUP BY) - Group Function (SUM, AVG, COUNT, MAX, MIN) - (ANY, ALL, EXISTS, NOT EXISTS)
Week 10	**Topic 6a: Logical Database Design (Part 1)** - Steps and Output in Logical Database Design - ERD to Relations
Week 11	**Topic 6b: Logical Database Design (Part 2)** - Normalization • Update Anomalies • Functional Dependencies • First, Second, and Third Normal Form • Boyce-Codd Normal Form

Week 12	**LAB: SQL- DML (Part 3)** - Joining Data from Multiple Tables • UNION, INTERSECT, EXCEPT • JOINs (Equi Join, Outer Join, Left Join)
Week 13	**Topic 7: Transaction Management** - Transaction Support - Key Characteristics of Transaction (ACID) - Concurrency Control in Transaction Management - Concurrency Control Method - Concurrency Control - Database Recovery **SQL 5a: Transaction Control Language (TCL)** - View - Transaction
Week 14	**Topic 8: Introduction to Data Warehouse** - Main concept of data warehouse - OLAP and OLTP - Architecture of data warehouse - Tools & Technologies (ETL) - Data Mart **SQL 5b: Transaction Control Language (TCL)** - Discretionary Access Control - Granting Privileges to Other Users (GRANT) - Revoking Privileges from Users (REVOKE)
Week 15	**Topic 9: Introduction to NOSQL Database**

Transferable skills (generic skills learned in course of study which can be useful and utilised in other settings):

Team working
Written communication

Faculty:	Computing	**Page:**	5 of 6
Program name:	Bachelor in Software Engineering with Honors		
Course code:	SCST1143	**Academic Session/Semester:**	2024/25/1
Course name:	Database	**Pre/co requisite (course name and code, if applicable):**	-
Credit hours:	3		

Student learning time (SLT) details:

Distribution of student Learning Time (SLT) Course content outline	Guided Learning (Face to Face)				Teaching and Learning Activities		TOTAL SLT
					Guided Learning Non-Face to Face	Independent Learning Non-Face to face	
CLO	**L**	**T**	**P**	**O**			
CLO 1	4h	3h			9h	33.5h	**49.5h**
CLO 2	24h	3h	3h		12h	4h	**46h**
CLO 3				5h		2h	**7h**
Grand Total SLT	**28h**	**6h**	**3h**	**5h**	**21h**	**39.5h**	**102.5h**

	Continuous Assessment	PLO (Code)	Percentage	Total SLT
1	Quiz(Take home)	PLO2 (CG)	5	**0.5h**
2	Lab Assignment	PLO3 (PS)	10	**6h**
3	Assignment	PLO2 (CG) PLO3 (PS)	10	**1h** **2h**
4	Test	PLO2 (CG) PLO3 (PS)	20	**2h**
5	PoBL Project	PLO3 (PS) PLO8 (LAR)	20 5	**3h** **As in CLO3 (2h)**
	Final Assessment		Percentage	Total SLT
1	Final Examination	PLO2 (CG) PLO3(PS)	30	**3h**
Grand Total			**100**	**120h**

L: Lecture, T: Tutorial, P: Practical, O: Others

***** To pass this course, students must pass both Continuous and Final assessment

Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):

Computer lab that is equipped with Database Management System (DBMS) software, ERD software designer tool, application program software

Learning resources:

Main references

Connolly,T., C.Begg. Database Systems: A Practical Approach to Design, Implementation, and Management. 13th Edition, Pearson Education International, 2019.

Additional references

Rob, P., Coronel, C. Database Systems: Design, Implementation, and Management. 10th Edition, Course Technology-Cengage Learning, 2012.

Online
http://elearning.utm.my

Academic honesty and plagiarism: *(Below is just a sample)*

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES). Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (Course policy, any specific instruction etc.):

1. Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.
2. Students are required to behave and follow the University's dressing regulation and etiquette all the time.
3. Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.
4. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not be accepted.
5. Make up exam will not be given, except to students who are sick and submit medical certificate confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.
6. To pass this course, students must pass both Continuous and Final assessment.

Disclaimer:

All teaching and learning materials associated with this course are for personal use only. The materials are intended for educational purposes only. Reproduction of the materials in any form for any purposes other than what it is intended for is prohibited.
While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.